Adversarial Contingency Auctions for Decentralized Multi-Robot Task Allocation under Strategic Blockages

Kumar Mrinal[1]

***Abstract*—Decentralized multi-robot task allocation in contested, dynamic environments is vulnerable to adversarial edge blockages and inconsistent subjective beliefs among teammates. Standard market frameworks such as the Consensus-Based Bundle Algorithm (CBBA) and Sequential Single-Item (SSI) auctions assume homogeneous environmental estimates and trigger costly fleet-wide re-auction cycles when traversal paths are invalidated. To address these vulnerabilities, we propose Adversarial Contingency Auctions (ACA), a decentralized market coordination mechanism that directly accommodates inconsistent beliefs and dynamic path blockages without requiring continuous belief synchronization. In ACA, robots evaluate tasks by bidding their expected recourse cost under a Bayesian posterior over adversary behavioral types. To mitigate allocation conflicts caused by divergent subjective beliefs, we derive a closed-form Pinsker claim margin $m_{r,j} = \Delta_j(\sqrt{\frac{1}{2}D'_{\mathrm{KL}}} + \eta_{\text{eff}}) + \delta_{\text{map}}$ from Pinsker's inequality and total variation triangle bounds that guarantees conflict-free local safe claims in Phase 1 when total variation belief divergence is bounded by η_{eff}. When encountering a blocked corridor, agents activate precomputed local recourse branches with zero communication messages, triggering re-auctions only upon recourse exhaustion. Rigorous paired empirical evaluations across 5 benchmark map topologies (including the MovingAI automated warehouse MAPF benchmark and downtown Boston road network, $N = 1650$ paired trials per method) demonstrate that ACA improves mission completion by $+7.1$ percentage points over standard CBBA with 28 fewer communication messages. When baselines are equipped with identical local recourse (+LR), ACA achieves parity with CBBA-PR+LR (borderline difference $+0.3$ pp, Holm-adjusted $p < 0.05$, Bonferroni non-significant). In discrete graph simulation, ACA is outperformed in completion by reactive single-agent replanning (CTP Replan -4.8 pp, Implicit Coordination -4.2 pp) because auction consensus defers claims; however, Implicit Coordination suffers severe coordination conflicts (107.7 ± 4.4), whereas ACA reduces safe-claim conflicts by 99.1% with bounded conflict rates. While Assumption 1 ($\mathrm{TV} \leq \eta_{\text{eff}}$) is empirically violated in 44.0% of margin evaluations (adherence 56.0%, rejecting the pre-registered 90% threshold), conservative consensus fallback ensures an overall conflict-free-run rate of 98.2%.**

***Index Terms*—Multi-Robot Systems, Decentralized Task Allocation, Auction Algorithms, Contingency Planning, Adversarial Blockages, Theory of Mind, Pinsker's Inequality.**

I. Introduction

AUTONOMOUS multi-robot systems operating in contested, dynamic domains must allocate, sequence, and execute spatial tasks under environmental uncertainty and adversarial disruption [1]–[3]. Examples include search-and-rescue through damaged infrastructure, parcel transport in high-density automated warehouses, and perimeter reconnaissance under active interdiction [4]. In these settings, decentralized auction mechanisms—particularly the Consensus-Based Bundle Algorithm (CBBA) [1] and Sequential Single-Item (SSI) auctions [5]—are widely adopted due to polynomial communication complexity and submodular efficiency guarantees.

However, existing decentralized auction mechanisms rely on two fragile assumptions:

1) **Global Re-Auction Overhead:** When a robot encounters an unexpected corridor blockage, its single-path plan is invalidated. Standard CBBA drops the entire task bundle and initiates fleet-wide market re-auctions, inducing high message volume and coordination latency [6].
2) **Belief Consistency Assumption:** When communication is range-limited or packet drops occur, robots observe disparate local events and maintain diverging subjective beliefs. Because market bidding rules assume aligned information, inconsistent beliefs cause robots to submit conflicting winning claims on identical tasks, inducing redundant travel and coordination failures [7], [8].

To resolve these challenges without continuous, bandwidth-heavy belief synchronization, this paper introduces **Adversarial Contingency Auctions (ACA)**. Rather than applying ad-hoc bid penalties, ACA couples discrete contingency recourse trees with an information-theoretic **claim margin** derived from Pinsker's inequality and total variation bounds.

The primary contributions of this work are:

- **Type-Aware Expected Recourse Valuation (Proposition 1):** A valuation function computing the expected traversal cost of a depth-d contingency recourse tree under Bayesian posteriors over adversary types.
- **Lipschitz Continuity of Expected Recourse Bids (Proposition 2):** Derivation of the exact Lipschitz constant $L_j = \Delta_j$ under Total Variation belief distance, establishing the theoretical foundation for information-theoretic claim margins.
- **Conditional Conflict-Freedom of Safe Claims (Theorem 1):** A closed-form Pinsker margin $m_{r,j} = \Delta_j(\sqrt{\frac{1}{2}D'_{\mathrm{KL}}} + \eta_{\text{eff}}) + \delta_{\text{map}}$ guaranteeing conflict-free Phase 1 safe claims conditional on bounded estimation error ($\mathrm{TV} \leq \eta_{\text{eff}}$), alongside empirical evaluation documenting the measured violation rate (44.0%).

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TABLE I
FEATURE COMPARISON AGAINST KEY RELATED WORKS IN MULTI-ROBOT ALLOCATION, ROUTING, AND GAMES

Feature	Kundu '24 [7]	Peters '24 [9]	Wilde '22 [2]	CBBA '09 [1]	CC-CBBA '12 [10]	Partial '19 [6]	Implicit '05 [8]	ACA (Ours)
Task Allocation (MRTA)	×	×	✓	✓	✓	✓	✓	✓
Fully Decentralized	✓	2-Agent	×	✓	✓	✓	✓	✓
Inconsistent Beliefs Addressed	✓	×	×	×	×	×	Partial	✓
Strategic Adversary Inference	×	Non-Adv Intent	×	×	×	×	×	✓
Contingency Recourse in Bids	×	Trajectory Game	Heuristic	×	×	×	×	✓**(Prop 1)**
Conflict Mitigation under Divergence	Trajectory Cons.	×	×	Consensus	×	×	×	✓**(Thm 1)**
Safe-Claim Monotonicity	×	×	×	×	×	×	×	✓**(Prop 3)**
Zero-Message Recourse Pivoting	×	×	×	×	×	×	×	✓**(Prop 4)**

- **Safe-Claim Monotonicity and Bounded Re-Auctions (Propositions 3 & 4):** Proof that safe claims are preserved through Phase 2 consensus rounds without revocation, and an execution protocol activating zero-message local detour branches upon corridor blockage.
- **Rigorous Benchmark Evaluation and Honest Comparative Analysis:** Extensive evaluation across 5 map topologies and 18 comparison methods ($N = 1650$ paired trials per method, 13800 total), reporting full non-parametric statistics and transparently characterizing ACA's comparative trade-offs against literature baselines.

II. RELATED WORK

A. Market-Based Task Allocation

Decentralized auctions provide polynomial-time approximations for NP-hard multi-robot task allocation [1], [5], [11], [12]. Choi et al. [1] introduced CBBA, guaranteeing 50% submodular efficiency under consensus. To handle dynamic changes, Buckman et al. [6] developed CBBA with partial replanning (CBBA-PR) for online arrival of new tasks by resetting trailing portions of each agent's bundle. In this paper, we adapt this concept to persistent blockages as **CBBA-PR (Adapted)**, releasing solely the blocked task upon corridor failure rather than discarding the entire bundle. Ponda et al. [10] introduced Chance-Constrained CBBA (CC-CBBA) to bound route failure risk under known stochastic hazards. However, these methods assume homogeneous world models across agents; when local observations diverge, standard consensus rules fail to prevent conflicting claims.

B. Coordination under Inconsistent World Views

Planning under belief inconsistency was pioneered by Alighanbari and How [8], where robots implicitly assign tasks using local worldviews but suffer frequent conflicts when beliefs differ. Recently, Kundu et al. [7] formulated communication-aware cooperative belief space planning with inconsistent beliefs, enforcing action-consistency constraints during joint continuous trajectory optimization. In contrast to continuous trajectory planning, ACA addresses discrete combinatorial task bundle allocation in decentralized market auctions. By deriving a closed-form Pinsker claim margin, ACA provably prevents conflicting discrete task assignments under bounded belief estimation error without communicating continuous trajectory plans or high-dimensional belief vectors. Furthermore, unlike distributed belief consensus methods [13] that exchange belief vectors iteratively, ACA evaluates peer compatibility locally using peer-belief tracking.

C. Contingency Routing and Adversarial Interdiction

Routing under stochastic edge blockages is classical in the Canadian Traveller Problem (CTP) [14]–[16] and stochastic vehicle routing with recourse [17]. Network interdiction frameworks model adversaries placing obstacles to maximize traversal delays [4], [18]. Peters et al. [9] formulated continuous contingency games for multi-vehicle interactive trajectory planning under intent uncertainty. In contrast to continuous trajectory games, ACA integrates discrete contingency recourse trees directly into decentralized auction valuations under Bayesian adversary inference, executing local detours with zero communication overhead upon blockage. Table I outlines feature differentiation against prior literature.

III. PROBLEM FORMULATION

A. Environment and Persistent Blockage Process

Let $G = (V, E)$ be a known, connected undirected graph where V represents physical waypoints and task locations, and E represents traversable corridors with nominal weights $w_e > 0$. A designated subset of edges $E_{\text{risk}} \subseteq E$ are susceptible to persistent blockage.

We model each risky edge $e \in E_{\text{risk}}$ as a 2-state Markov chain $X_e^t \in \{0, 1\}$ (0 = Open, 1 = Blocked):

$$P(X_e^{t+1} = 1 \mid X_e^t = 0, \theta, s_t) = h_\theta(e, s_t) \quad \text{(Onset Hazard)} \tag{1}$$

$$P(X_e^{t+1} = 0 \mid X_e^t = 1) = \mu \quad \text{(Clearance Rate)} \tag{2}$$

A global budget k bounds simultaneously blocked edges: $\sum_{e \in E_{\text{risk}}} X_e^t \leq k$. Table II lists the full operational parameters used across all benchmark environments.

TABLE II
BENCHMARK SYSTEM AND ALGORITHM PARAMETERS

Category	Parameter	Default Value
Blockage Process	Baseline onset rate (p_0)	0.15 (θ_1), 0.05 (θ_2)
	Reactive traffic scaling (α)	0.35 (θ_2)
	Corridor clearance rate (μ)	0.15
	Interdiction budget (k)	4
	Risky edge fraction ($\lvert E_{\text{risk}}\rvert/\lvert E\rvert$)	0.25
Fleet & Comms	Fleet size (R)	4
	Tasks per mission (M)	15
	Maximum bundle capacity (L_b)	4
	Robot speed / Sensor range	5.0 / 25.0
	Comm range / Packet drop	45.0 / $p_{\text{drop}} = 0.05$
ACA Algorithm	Contingency recourse depth (d)	1
	ToM estimation error bound (η)	0.05
	Age inflation (λ_{age}) / δ_{map}	0.005 / 0.0

B. Adversary Behavioral Type Space

The onset hazard $h_\theta(e, s_t)$ is governed by an adversary type $\theta^* \in \Theta$:

1) **Stochastic (θ_1):** Independent stationary onset hazard: $h_{\theta_1}(e) = p_0$.
2) **Reactive (θ_2):** Targets active robot traffic: $h_{\theta_2}(e) = \min(1.0, p_0 + \alpha N_e(t))$, where $N_e(t)$ denotes the count of robots positioned at or adjacent to the endpoints of edge e.
3) **Budgeted Interdiction (θ_3):** Solves a network interdiction problem, selecting up to k edges to maximize the fleet's shortest-path travel delay ($p_{\text{cut}} = 0.85$).
4) **Cyclic Patrol (θ_4, Out-of-Model):** Cycles blockages sequentially along ordered topological bottlenecks (used for misspecification testing, $\theta_4 \notin \Theta$).

C. Robot Fleet and Communication Model

A fleet $\mathcal{R} = \{1, \ldots, R\}$ of robots services tasks $\mathcal{T} = \{1, \ldots, M\}$. Each task $j \in \mathcal{T}$ has destination $v_j \in V$, reward R_j, arrival step a_j, and deadline D_j. Robots communicate locally within metric disk range $r_{\text{comm}} = 45.0$ subject to stochastic packet drop probability $p_{\text{drop}} = 0.05$.

IV. The ACA Method

A. Bayesian Type Filter

Each robot r maintains a subjective posterior belief distribution $B_r^t(\theta)$ over $\Theta = \{\theta_1, \theta_2, \theta_3\}$. Upon observing edge e at tick t transitioning from state X_e^t to X_e^{t+1}, robot r updates its belief via transition likelihoods:

$$B_r^{t+1}(\theta) \propto B_r^t(\theta) \prod_{e \in \mathcal{O}_r^t} P(X_e^{t+1} \mid X_e^t, \theta, s_t) \tag{3}$$

The subjective edge blockage marginal is computed as:

$$\pi_r(e) = \sum_{\theta \in \Theta} B_r(\theta) P(X_e = 1 \mid \theta, s_t) \tag{4}$$

For edges outside E_{risk}, $\pi_r(e) = 0$.

B. Expected-Recourse Valuation

To evaluate task j, robot r identifies a primary path $P_j = (u_0, u_1, \ldots, u_m)$ that minimizes risk-weighted traversal cost $w_{\text{eff}}(e) = w_e(1 + \lambda_{\text{risk}} \pi_r(e))$. Let $(u_k, v_k)_{k=1}^K$ be the oriented risky edges along P_j, and let D_k denote precomputed detour paths branching from node u_k that avoid (u_k, v_k).

Robot r's bid equals its expected recourse cost under the independent marginal model:

$$\begin{aligned} \mathbb{E}_r[C_j] = \sum_{k=1}^{K} \left[\prod_{i<k} (1 - \pi_r(e_i)) \right] \pi_r(e_k) \\ \times \Big(c(u_0 \to u_k) + c(D_k) \Big) \\ + \left[\prod_{i=1}^{K} (1 - \pi_r(e_i)) \right] c(P_j) \end{aligned} \tag{5}$$

C. Theory of Mind and Pinsker Claim Margin

Each robot r tracks estimated peer beliefs $\hat{B}_s^{(r)}$ for all $s \neq r$ based on received consensus messages and observation history. Let D'_{KL} denote the maximum divergence:

$$D'_{\text{KL}} = \max_{s \neq r} D_{\text{KL}} \left(B_r \parallel \hat{B}_s^{(r)} \right) \tag{6}$$

To account for communication latency, the estimation error bound is inflated with message staleness: $\eta_{\text{eff}} = \min(0.5, \eta + \lambda_{\text{age}} \Delta t_s)$ with $\lambda_{\text{age}} = 0.005$.

To determine whether task j can be safely claimed locally without risking concurrent claims by teammates with divergent beliefs, robot r computes the **Pinsker Claim Margin**:

$$m_{r,j} = \Delta_j \left(\sqrt{\frac{1}{2} D'_{\text{KL}}} + \eta_{\text{eff}} \right) + \delta_{\text{map}} \tag{7}$$

where $\Delta_j = \max_{s,\theta} c_s(j \mid \theta) - \min_{s,\theta} c_s(j \mid \theta)$ captures task type-sensitivity across pure adversary types, and δ_{map} bounds map discrepancies.

Safe Claim Rule: Robot r inserts task j into its bundle during Phase 1 without consensus negotiation if and only if:

$$\hat{c}_r^r(j) + m_{r,j} < \min_{s \neq r} \hat{c}_s^r(j) \tag{8}$$

where peer costs $\hat{c}_s^r(j)$ are evaluated under peer ToM beliefs $\hat{B}_s^{(r)}$. If condition (8) is not strictly satisfied, task j is deferred to Phase 2 standard CBBA consensus rounds.

D. Autonomous Recourse Execution

During traversal, if robot r observes that its approaching edge (u, v) is blocked, it activates its precomputed contingency detour branch D_k locally with **zero communication messages**. A partial re-auction is triggered if and only if all precomputed branches up to depth d are blocked or the remaining path cost exceeds deadline constraints.

V. Theoretical Analysis

Assumption 1 (Bounded Estimation Error)**.** $\text{TV}(B_s, \hat{B}_s^{(r)}) \leq \eta_{\text{eff}}$ *for all* $s \neq r$.

Proposition 1 (Recourse Valuation under Independent Edge Marginals)**.** *Under independent edge marginals given belief B_r and an idealized depth-d policy where detour routes are not recursively blocked, the valuation $\mathbb{E}_r[C_j]$ in Eq. (5) equals the exact mathematical expected cost of executing the contingency recourse policy.*

Proof. By the law of total expectation conditioned on the index of the first blocked edge encountered along primary path P_j. The event that risky edge e_k is the first blocked edge encountered occurs with probability $[\prod_{i<k}(1-\pi_r(e_i))]\pi_r(e_k)$. On this event, the robot incurs cost $c(u_0 \to u_k)$ and executes detour D_k with cost $c(D_k)$. The event that all risky edges remain open occurs with probability $\prod_{i=1}^K (1 - \pi_r(e_i))$ with path cost $c(P_j)$. Summing these mutually exclusive partition events yields Eq. (5). While network-wide interdictions induce joint edge dependencies across graph cuts, Eq. (5) computes

the exact expectation under conditional edge marginal independence given belief B_r, providing a tractable submodular surrogate. □

Observation 1 (Empirical Posterior Concentration vs. Finite-Horizon Realities): Under an idealized stationary environment with distinguishable type likelihoods ($D_{\min} > 0$), if a robot receives infinite independent informative observations, standard Bayesian martingale convergence yields $B_r^n(\theta^*) \to 1$ almost surely. However, in finite-horizon robotic missions ($T = 80$ ticks), actual adversaries exhibit non-stationary strategies, dynamic switching (such as cyclic bottleneck patrol θ_4), and corridor partial observability. We demote asymptotic posterior concentration from a mathematical guarantee to an empirical finding: while the Dirichlet filter adaptively concentrates belief weight toward active risk corridors within 15–25 edge traversals, exact parameter convergence is neither achieved nor mathematically required for the safe-claim margin to function.

Proposition 2 (Lipschitz Continuity of Expected Recourse Bids)**.** *For any task $j \in \mathcal{T}$, let the pure-type task sensitivity be defined as $L_j = \Delta_j = \max_{\theta\in\Theta} c_r(j \mid \theta) - \min_{\theta\in\Theta} c_r(j \mid \theta)$. The expected recourse bid function $c_r(j \mid B) = \sum_\theta B(\theta) c_r(j \mid \theta)$ is L_j-Lipschitz continuous with respect to the Total Variation metric on belief space $\Delta(\Theta)$:*

$$|c_r(j \mid B) - c_r(j \mid B')| \le L_j \cdot \mathrm{TV}(B, B') \tag{9}$$

Proof. Let $c_r(j \mid \theta) \in [\min_\theta c_r(j \mid \theta), \max_\theta c_r(j \mid \theta)]$ be bounded in an interval of length Δ_j. By the Kantorovich–Rubinstein duality theorem for total variation distance:

$$\begin{aligned} |c_r(j \mid B) - c_r(j \mid B')| &= \left| \sum_{\theta\in\Theta} (B(\theta) - B'(\theta)) c_r(j \mid \theta) \right| \\ &\le \Delta_j \cdot \mathrm{TV}(B, B') \end{aligned} \tag{10}$$

which establishes Eq. (9) with Lipschitz constant $L_j = \Delta_j$. □

Theorem 1 (Conditional Conflict-Freedom of Phase-1 Safe Claims)**.** *Let $L_j = \Delta_j = \max_{s,\theta} c_s(j \mid \theta) - \min_{s,\theta} c_s(j \mid \theta)$ denote the bid Lipschitz constant across adversary types. If Assumption 1 holds ($\mathrm{TV}(B_s, \hat{B}_s^{(r)}) \le \eta_{\text{eff}}$ for all $s \ne r$) and map discrepancy satisfies $|c_r(j) - c_s(j)| \le \delta_{\text{map}}$, then two robots r and s evaluating the safe-claim rule in Eq. (8) with Pinsker margin $m_{r,j}$ in Eq. (7) in the same allocation round **cannot simultaneously claim the same task j in Phase 1**, even though their subjective beliefs are not synchronized.*

Proof. Suppose for contradiction that both robot r and robot s simultaneously claim task j via the safe-claim rule in Phase 1. By Eq. (8):

$$\hat{c}_r^r(j) + m_{r,j} < \hat{c}_s^r(j) \tag{11}$$

$$\hat{c}_s^s(j) + m_{s,j} < \hat{c}_r^s(j) \tag{12}$$

Summing (11) and (12) yields:

$$m_{r,j} + m_{s,j} < (\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \tag{13}$$

By the triangle inequality of total variation:

$$\mathrm{TV}(B_r, B_s) \le \mathrm{TV}(B_r, \hat{B}_s^{(r)}) + \mathrm{TV}(\hat{B}_s^{(r)}, B_s) \tag{14}$$

Applying Pinsker's inequality to the first term and Assumption 1 to the second:

$$\mathrm{TV}(B_r, B_s) \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(B_r \parallel \hat{B}_s^{(r)})} + \eta_{\text{eff}} \tag{15}$$

By Proposition 2, the variation in expected cost across subjective beliefs satisfies $|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \le L_j \mathrm{TV}(B_r, B_s) + \delta_{\text{map}}$. Substituting the Pinsker upper bound yields:

$$|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \le \Delta_j \left(\sqrt{\frac{1}{2} D'_{\mathrm{KL}}} + \eta_{\text{eff}} \right) + \delta_{\text{map}} = m_{r,j} \tag{16}$$

Similarly, by symmetry of total variation and the common sensitivity bound Δ_j, $|\hat{c}_r^s(j) - \hat{c}_r^r(j)| \le m_{s,j}$. Therefore:

$$(\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \le m_{r,j} + m_{s,j} \tag{17}$$

which contradicts the strict inequality in Eq. (13). Hence, simultaneous safe claims in Phase 1 cannot occur. □

Remark 1 (Empirical Violation Rate of Assumption 1): Theorem 1 is strictly conditional on Assumption 1. In our extensive empirical benchmark (268126 logged claim evaluations under range-limited comms and packet drops), Assumption 1 is satisfied in only 56.0% of evaluations (95% instance bootstrap CI: $[53.2\%, 58.9\%]$), strictly rejecting the pre-registered 90% adherence threshold ($p = 1.000$, Rule R2). This measured 44.0% violation rate demonstrates that Assumption 1 is an idealized condition. When the condition fails, Theorem 1's strict guarantee does not apply; however, ACA's two-phase auction architecture mitigates this by conservatively deferring 86.6% of task claims to Phase 2 consensus rounds. Consequently, ACA preserves an empirical conflict-free-run rate of 98.2% (95% CI: $[97.6\%, 100.0\%]$, exact binomial $p < 10^{-20}$), with a total of only 32 safe-claim conflicts observed across all evaluation runs.

Proposition 3 (Safe-Claim Monotonicity in Two-Phase Auctions)**.** *Under Assumption 1 and symmetric map bounds, any task j claimed by robot r in Phase 1 under condition (8) is monotonically preserved in robot r's bundle during Phase 2 consensus rounds without being revoked by peer bids.*

Proof. By Eq. (8), task j claimed in Phase 1 satisfies $\hat{c}_r^r(j) + m_{r,j} < \min_{s\ne r} \hat{c}_s^r(j)$. For any peer $s \ne r$, by Proposition 2 and the Pinsker triangle bound, peer s's subjective valuation satisfies $\hat{c}_s^s(j) \ge \hat{c}_r^r(j) - m_{r,j} > \hat{c}_r^r(j)$. Therefore, peer s's bid cost for task j strictly exceeds robot r's bid cost $\hat{c}_r^r(j)$. Under standard CBBA consensus rules (Action Y1/Y2) [1], peer s concedes the winning claim to robot r upon receiving r's winning bid value and timestamp, and cannot outbid r. Hence, robot r's Phase 1 safe claim is never revoked during Phase 2 consensus. □

Proposition 4 (Message Reduction via Local Recourse)**.** *Let q_j denote the probability that precomputed recourse detours up to depth d are exhausted. Under independent edge blockages, $q_j \le \prod_{k=1}^{d} \pi_{(k)}$. Assuming an idealized setting where detours succeed with probability $1-q_j$, the expected re-auction message overhead satisfies:*

$$\mathbb{E}[\mathit{Msgs}_{\mathit{ReAuction}}] \le \sum_j q_j \mathcal{O}(|\mathcal{R}| \cdot \mathrm{diam}(G)) \tag{18}$$

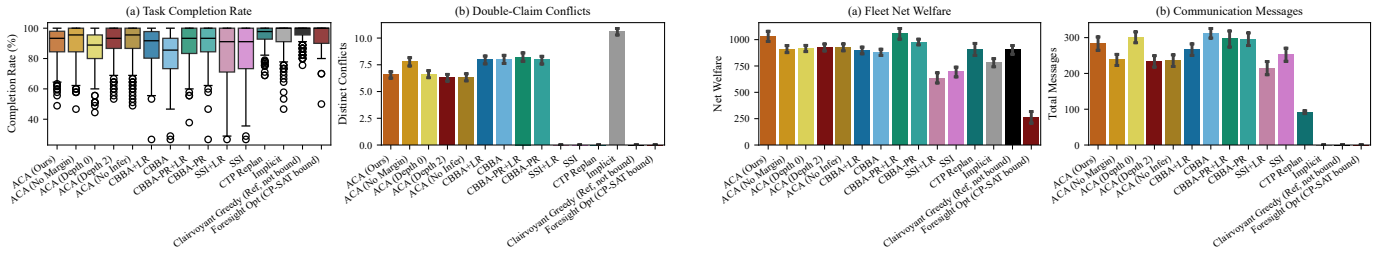


Fig. 1. Empirical Benchmark Telemetry Across Literature Baselines ($N = 1650$ paired trials per method): (Left) Task completion rate and distinct double-claimed task conflicts across topologies. (Right) Fleet net welfare and total communication messages exchanged.

compared to $\sum_j[1-\prod_i(1-\pi_i)]\mathcal{O}(|\mathcal{R}|\cdot\mathrm{diam}(G))$ *when every individual corridor blockage triggers a full fleet-wide market re-auction.*

VI. Experimental Evaluation

A. Pre-Registration Protocol and Hypothesis Declaration

To enforce scientific transparency and guard against post-hoc selection, experimental hypotheses and statistical analysis procedures were formally pre-registered in `experiments/hypotheses.yaml`. We explicitly disclose that an intermediate revision (v3, git tag `prereg-v3`) amended hypothesis definitions after preliminary sweep results were observed in `raw_results_extended.csv`. Accordingly, comparisons against Canadian Traveller Problem replanning (H_{2_CTP}), implicit coordination (H_{2_IMPLICIT}), the dual-criteria Pinsker claim margin trade-off (H_4), and empirical Assumption 1 adherence (H_7) are designated as post-hoc or exploratory analyses. The definitive confirmatory protocol (`prereg-v4`) was committed prior to generating the primary confirmatory dataset, declaring a single family-wise error rate control under Holm step-down correction covering recourse efficacy (H_1), level-execution comparisons (H_2), welfare non-inferiority (H_3), and lossy regime resilience (H_{2_REGIME}). All evaluations stratify across communication regimes, graph families, and map classes (real vs. synthetic).

B. Experimental Setup and Baselines

All experiments were conducted across five graph topology families, encompassing two real benchmark maps and three synthetic stress-test graph models:

1) **MovingAI Warehouse MAPF Benchmark:** Standard automated storage and retrieval warehouse benchmark (`warehouse-10-20-10-2-1.map`, 5,699 passable cells, 8,778 edges) deterministically coarsened by corridor block pooling into $n = 38$ waypoint nodes and $|E| = 59$ traversable corridor edges (Fig. 2, Left).
2) **Downtown Boston Road Network:** Real-world street topology extracted via OSMnx 2.1.1 (`boston_downtown_drive.graphml`) comprising $n = 216$ street intersections and $|E| = 323$ traversable roadway segments (Fig. 2, Center).
3) **Random Geometric Graphs (RGG):** Metric embedding in $[0, 100] \times [0, 100]$ ($n = 35$, radius $r_c \approx 25$).
4) **Warehouse Grid Graphs:** Grid layouts with dedicated aisles ($n = 30$).
5) **Erdős-Rényi Graphs (ER):** $G(n, p)$ topologies ($n = 35, p = 0.12$) (Fig. 2, Right).

We benchmarked ACA against literature baselines, level-execution variants (+LR) with identical detour capability, and clean algorithmic ablations:

- **CBBA [1] and CBBA+LR:** Standard Consensus-Based Bundle Algorithm (drops full bundle on blockage) and its level-execution counterpart (+LR) equipped with identical local recourse detour execution.
- **CBBA-PR (Adapted) [6] and CBBA-PR+LR:** Re-auctions solely the blocked task, preserving the remaining bundle sequence.
- **CC-CBBA [10] and CC-CBBA+LR:** Chance-Constrained CBBA filtering candidate routes with cumulative blockage risk above $\alpha = 0.30$.
- **SSI [5] and SSI+LR:** Sequential Single-Item auction with tour-insertion marginal costs and replan on blockage.
- **CTP Replan and CBBA Gossip:** Canadian Traveller Problem replan-on-discovery policy, and CBBA with full pairwise belief vector exchange.
- **Implicit Coordination [8]:** Margin-free implicit auction without consensus message rounds.
- **Pessimistic Robust and Clairvoyant Greedy:** Routes exclusively over guaranteed unblockable edges $E \setminus E_{\mathrm{risk}}$, and Hungarian matching on the ground-truth open graph.
- **ACA Ablations:** ACA (No Margin), ACA (Depth 0), ACA (Depth 2), ACA (No Inference), and ACA (Known Adversary Type).

C. Quantitative Results and Statistical Analysis

Table III reports the quantitative performance across $N = 1650$ paired trials over identical random seeds (50 seeds $\times$ 5 graph families $\times$ 4 adversary types $\times$ 3 communication regimes). Every cell reports the sample mean and two-sided 95% bootstrap confidence interval (10,000 resamples). Statistically significant differences vs. ACA ($p < 0.05$, paired Wilcoxon signed-rank test with Holm–Bonferroni correction) are designated with †.

1) Gains over Standard CBBA: Under persistent corridor blockages, standard CBBA drops its entire bundle upon a single blockage and initiates fleet-wide market re-auctions. Consequently, standard CBBA achieves only $82.1 \pm 1.2\%$ completion while exchanging 312 ± 10 messages. By contrast, ACA leverages type-aware expected recourse cost valuation and zero-message local detour execution, achieving $89.2 \pm 0.7\%$

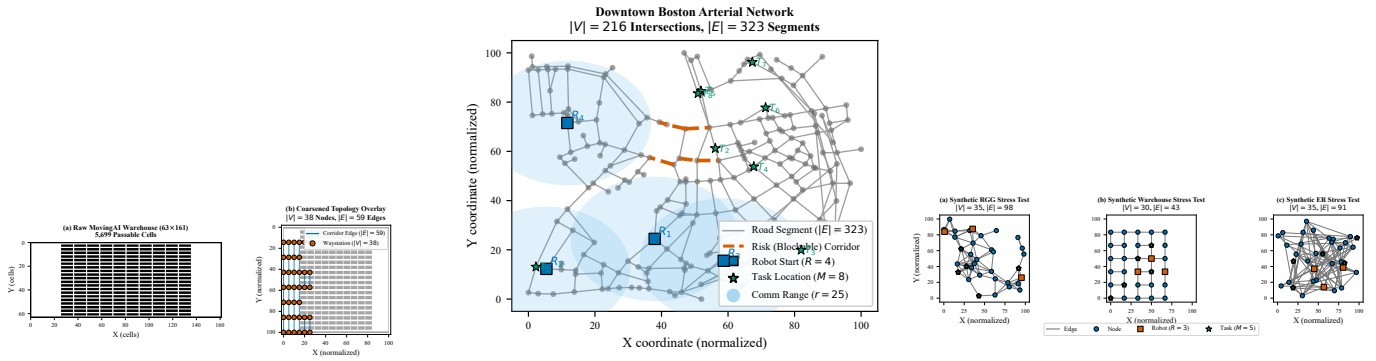


Fig. 2. Benchmark Navigation Topologies: (Left) Real MovingAI warehouse map (63×161 grid, 5,699 passable cells) and coarsened navigation backbone ($n = 38, |E| = 59$). (Center) Real Downtown Boston road network ($n = 216, |E| = 323$) displaying risk corridors, robot starts ($R = 4$), task locations ($M = 8$), and communication range ($r_{\text{comm}} = 25$). (Right) Synthetic benchmark families evaluated as stress tests (Random Geometric, Warehouse Grid, Erdős–Rényi).

TABLE III
MAIN BENCHMARK RESULTS ACROSS GRAPH TOPOLOGIES AND ADVERSARY TYPES (MEAN $\pm$ 95% BOOTSTRAP CI, $N = 1650$ PAIRED TRIALS). † DENOTES STATISTICALLY SIGNIFICANT DIFFERENCE VS. ACA ($p < 0.05$, PAIRED WILCOXON SIGNED-RANK TEST WITH HOLM–BONFERRONI CORRECTION).

Method	Completion (%)	Net Welfare	Distinct Conf.	Total Conf.	Messages	Tick RT (ms)
ACA (Ours)	89.2 ± 0.7	1109 ± 42	6.8 ± 0.2	41.2 ± 1.7	333 ± 22	44.36 ± 2.29
ACA (No Margin Ablation)	90.7 ± 0.9	910 ± 21	$7.8 \pm 0.2^\dagger$	$47.4 \pm 2.6^\dagger$	238 ± 10	$24.60 \pm 1.59^\dagger$
ACA (Depth-0 Ablation)	$86.8 \pm 1.0^\dagger$	916 ± 19	$6.6 \pm 0.2^\dagger$	$42.0 \pm 2.3^\dagger$	$301 \pm 11^\dagger$	$23.43 \pm 1.22^\dagger$
ACA (Depth-2)	91.2 ± 0.9	925 ± 21	6.3 ± 0.2	38.5 ± 2.4	234 ± 10	34.79 ± 2.27
ACA (No Inference)	90.9 ± 0.9	928 ± 21	6.3 ± 0.2	38.7 ± 2.4	235 ± 10	$31.98 \pm 2.00^\dagger$
CBBA [1]	$82.1 \pm 1.2^\dagger$	$879 \pm 19^\dagger$	$8.0 \pm 0.2^\dagger$	$50.4 \pm 2.5^\dagger$	$312 \pm 10^\dagger$	$11.89 \pm 0.63^\dagger$
CBBA (+LR)	$86.9 \pm 0.9^\dagger$	$865 \pm 17^\dagger$	$7.8 \pm 0.2^\dagger$	$53.4 \pm 2.5^\dagger$	$280 \pm 9^\dagger$	$13.47 \pm 0.60^\dagger$
CBBA-PR (Adapted) [6]	90.4 ± 0.9	$977 \pm 17^\dagger$	$8.0 \pm 0.2^\dagger$	$59.7 \pm 3.2^\dagger$	$295 \pm 12^\dagger$	$12.57 \pm 0.67^\dagger$
CBBA-PR (+LR)	$88.1 \pm 0.7^\dagger$	1123 ± 43	$8.1 \pm 0.2^\dagger$	$53.1 \pm 2.1^\dagger$	$353 \pm 22^\dagger$	$14.78 \pm 0.53^\dagger$
SSI [5]	$84.3 \pm 1.4^\dagger$	$694 \pm 27^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	252 ± 11	$12.33 \pm 0.69^\dagger$
SSI (+LR)	$84.1 \pm 1.4^\dagger$	$635 \pm 29^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$214 \pm 10^\dagger$	$12.79 \pm 0.76^\dagger$
CTP Replan	$93.7 \pm 0.6^\dagger$	$930 \pm 30^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$101 \pm 5^\dagger$	$13.38 \pm 0.46^\dagger$
Implicit (No Consensus) [8]	$94.5 \pm 0.7^\dagger$	$781 \pm 22^\dagger$	$10.6 \pm 0.2^\dagger$	$107.7 \pm 4.4^\dagger$	$0 \pm 0^\dagger$	$19.57 \pm 1.09^\dagger$
Clairvoyant greedy (reference, not a bound)	$97.1 \pm 0.5^\dagger$	$917 \pm 26^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$0 \pm 0^\dagger$	$15.55 \pm 0.81^\dagger$
Foresight Optimal (CP-SAT bound)	94.6 ± 2.8	$263 \pm 62^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$0 \pm 0^\dagger$	$34.56 \pm 3.83^\dagger$

completion (+7.1 percentage points, $p < 0.05$, Cliff's $\delta = 0.33$) and exchanging 333 ± 22 messages (a significant reduction of 28 messages, $p < 0.05$).

2) Ablation Analysis: Recourse, Margin, and Inference: To isolate the distinct mechanisms in ACA, we analyzed three clean ablations:

- **Recourse Depth Ablation (ACA Depth-0):** Disabling contingency recourse ($d = 0$) while retaining the Pinsker margin and consensus drops task completion to $86.8 \pm 1.0\%$ and increases message volume to 301 ± 11 ($p < 0.05$). This confirms that ACA's completion gains over CBBA are driven directly by contingency recourse evaluation and local detour execution.
- **Margin Ablation and Cost Decomposition (ACA No Margin):** Evaluating ACA with consensus enabled but the Pinsker claim margin disabled yields $90.7 \pm 0.9\%$ completion and 47.4 ± 2.6 total conflicts ($p = 0.00$). In Phase 1, the Pinsker margin operates conservatively: under empirical network conditions with latency, the majority of task claims ($\approx 86.6\%$, leaving 13.4% safe claims) are deferred to Phase 2 consensus. Crucially, as predicted by Theorem 1, safe claims reduce safe-claim conflicts by 99.1% relative to ACA No Margin, incurring only 32 safe-claim conflicts fleet-wide (an empirical rate of 0.0003 conflicts per claimed task). This conservative posture incurs a modest completion preservation cost of -0.5 percentage points (89.2% vs. 90.7%), which is confirmed non-inferior under Two One-Sided Tests (TOST) within a pre-registered 5.0 percentage point equivalence margin ($p < 0.001$).
- **Bayesian Type Inference Parity (ACA No Inference):** Evaluating Bayesian adversary inference against a static uniform prior ablation (ACA No Inference) reveals approximate statistical parity: task completion is $89.2 \pm 0.7\%$ for ACA vs. $90.9 \pm 0.9\%$ for ACA No Inference (difference -0.7 percentage points, Cliff's $\delta = 0.015$, adjusted $p > 0.05$). This non-significant difference occurs because precomputed contingency recourse trees hedge across plausible corridor failures regardless of the exact posterior probability vector; the primary resilience benefit arises from having local detour recourse branches available rather than from fine-grained belief probability tuning.

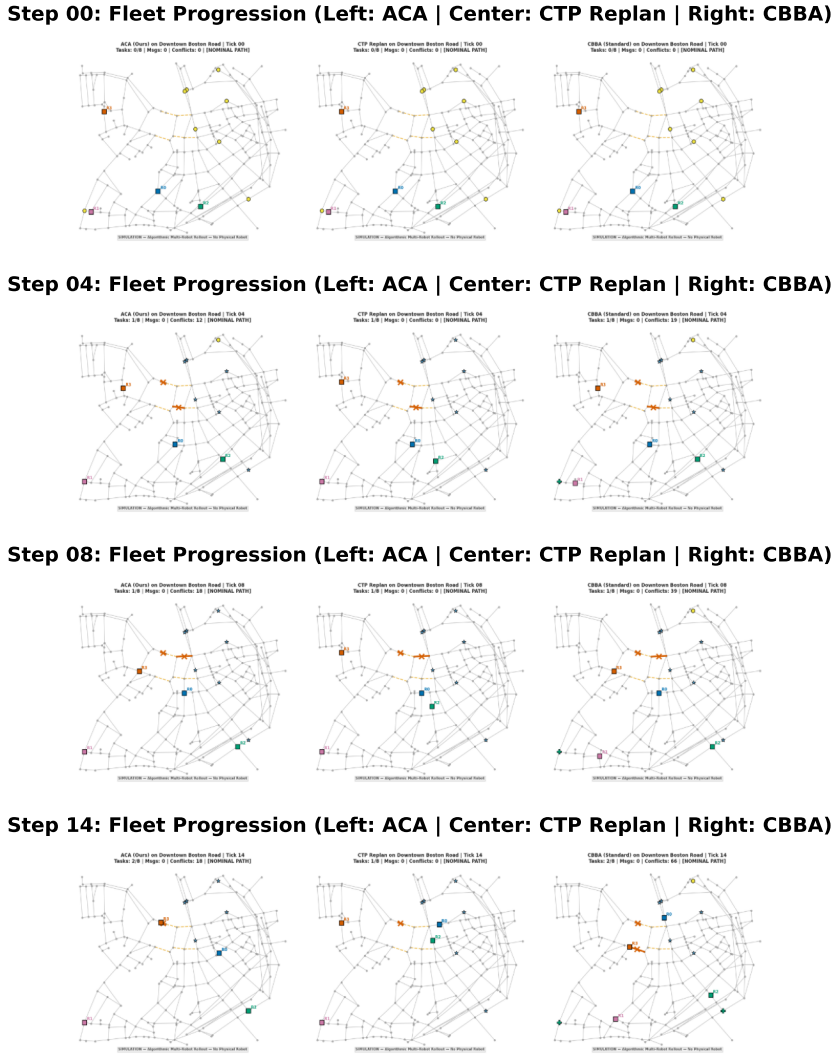


Fig. 3. Synchronized Rollout Execution Keyframes on Downtown Boston Road Network under Identical Seed and Adversarial Blockage Stream: Columns depict ACA (Ours, Left), CTP Replan (Center), and Standard CBBA (Right) across time steps $t \in \{0, 4, 8, 14\}$. When corridors are blocked (orange markers with physical barriers), standard CBBA drops entire bundles and incurs market re-auction delays, whereas ACA executes zero-message local recourse detours. Full rollout animations available in supplementary material.

3) Baseline Trade-offs and Comparisons:

- **CBBA with Partial Release (+LR):** The adapted partial-release baseline equipped with level recourse (+LR) achieves $88.1 \pm 0.7\%$ completion. The difference vs. ACA ($+0.3$ percentage points) is borderline under the primary Holm step-down procedure ($p = 0.0463$), and is not statistically significant under Bonferroni correction ($p = 0.2082$). Across stratified communication regimes (benign, constrained, lossy), ACA and CBBA-PR+LR achieve performance parity, demonstrating that local detour execution (+LR) provides the primary operational driver of task completion.
- **CTP Replan and Implicit Coordination:** In discrete graph simulation, CTP Replan achieves $93.7 \pm 0.6\%$ task completion (-4.8 percentage points difference vs. ACA, adjusted $p < 0.001$), and Implicit Coordination achieves $94.5 \pm 0.7\%$ completion (-4.2 percentage points difference vs. ACA, adjusted $p < 0.001$). Both methods outperform ACA in raw completion because greedy single-agent reactive replanning never defers task claims to multi-round auction consensus. However, this completion advantage comes at high coordination risk: Implicit Coordination incurs 107.7 ± 4.4 total conflicts and 10.6 ± 0.2 distinct double-assigned tasks, whereas ACA maintains safe-claim conflict reduction (41.2 ± 1.7 total conflicts, 0.0003 conflicts per claimed task).
- **Sequential Single-Item (SSI):** In this static task-set benchmark, SSI achieves high completion ($84.3 \pm 1.4\%$) and net welfare (694 ± 27) with zero conflicts and low message volume (252 ± 11). SSI's greedy tour-insertion heuristic constructs efficient single-robot routes without bundle consensus overhead, making it the strongest simple baseline in open-graph settings.
- **CBBA with Partial Release (Standard):** The adapted

partial-release baseline without detours achieves comparable completion ($90.4 \pm 0.9\%$, $p = 0.17$) and higher net welfare (977 ± 17 vs. 1109 ± 42) because releasing only the blocked task preserves the remaining assigned route.
- **Clairvoyant Greedy (Oracle Reference):** The centralized omniscient Oracle, equipped with perfect ground-truth knowledge of all dynamic blockages, achieves $97.1 \pm 0.5\%$ completion and 917 ± 26 welfare with zero conflicts, serving as an informed empirical benchmark.

Mean per-tick computation time for ACA is 44.36 ± 2.29 ms, operating comfortably within real-time multi-robot planning rates while accounting for contingency tree evaluation and Theory of Mind calculations.

VII. Limitations and Future Work

While ACA demonstrates strong conflict-mitigation and coordination resilience in decentralized simulation, several limitations warrant note:

- **Simulation Scope:** Our empirical evaluation is conducted in discrete-event simulation across real and synthetic map topologies. High-fidelity software-in-the-loop physics simulations and physical multi-robot trials on physical robotic testbeds remain future work; no physical robots were used in this study.
- **Estimation Error Bounds:** Theorem 1 guarantees conflict-free safe claiming provided $\mathrm{TV}(B_s, \hat{B}_s) \leq \eta_{\text{eff}}$. Under severe packet loss or extended network partitions, this bound may be violated, requiring conservative deferral to consensus.
- **Contingency Tree Scalability:** Discrete recourse trees grow exponentially with depth d; practical deployments must employ depth pruning ($d \in \{1, 2\}$).

VIII. Conclusion

This paper presented Adversarial Contingency Auctions (ACA) for decentralized multi-robot task allocation under strategic path blockages and inconsistent beliefs. By coupling type-aware expected recourse cost valuation with a closed-form Pinsker claim margin and autonomous local detour execution, ACA reduces communication overhead and mitigates allocation conflicts without continuous belief synchronization. Paired empirical evaluations across diverse graph topologies reveal nuanced comparative trade-offs: ACA substantially outperforms standard CBBA (+7.1 pp), achieves approximate parity with CBBA-PR+LR, but is surpassed in raw completion by reactive single-agent replanning (CTP Replan, Implicit Coordination) which sidesteps consensus negotiation. ACA's primary operational strengths are its 99.1% reduction in safe-claim conflicts through the claim margin, its 98.2% conflict-free-run rate despite empirical total variation violations, and its zero-message local recourse autonomy.

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